

Lab 9.4 Machine Learning 2 - Machine Learning Implementation to Edge Device

Working with MySQL Database and Grafana Dashboard

4.1 MySQL Database

Let's move on to the data aggregation for the MySQL database. Recall [Lab6](#) and [Lab7](#) for this part. The information of the MySQL server and database is below.

- DNS: mepotrb16.ecn.purdue.edu
- Port: 3306
- Account: firstnamelastname (e.g., johndoe)
- Database: ME597Spring26
- Table: firstnamelastname_lab6 (e.g., johndoe_lab6)
 - Reuse the Table you created in lab6.

The data collector Python script ([mtconnect_collector.py](#)) using XML parsing is prepared for you as Lab7. Follow the steps below to collect all data stream from the MTConnect agent.

1. Run the sample data collector Python script ([mtconnect_collector.py](#)) on either laptop or Raspberry Pi to store all data stream from the MTConnect agent.
- a. You need to modify some front parts that are wrapped in asteriks below. 2. Check if data is being stored using 'MySQL Workbench' on laptop.

Python - Python3 ([mtconnect_collector.py](#))

```
# Credential
*HOST* = 'MySQL HOST DNS' # MySQL server host DNS
*PORT* = 3306 # MySQL server port number
*USER* = 'your account' # MySQL account name
*PASSWORD* = 'your password' # Password of the account
*DB* = 'Database name' # DB name
*TABLE* = 'your table name' # table name
## Credential

## MTConnect info.
*agent* = "agent host"
*agent_port* = "agent port number"
url_current = "http://" + agent + ":" + agent_port + "/current"
MTCONNECT_STR = ET.fromstring(requests.get(url_current).content).tag.split("}")[0] + "}"
print("MTConnect string header is {}".format(MTCONNECT_STR))
## MTConnect info
```

JohnDoe_lab6 x

Limit to 1000 rows

1 • SELECT * FROM ME597Spring24.johndoe_lab6;

Result Grid

id	timestamp	sensor	measurement	value
187	2022-03-31 16:34:30	ADXL345	mae	0.029508354
188	2022-03-31 16:34:31	ADXL345	mae	0.029577313
189	2022-03-31 16:34:32	ADXL345	mae	0.032998398
190	2022-03-31 16:34:34	ADXL345	mae	0.037064757
191	2022-03-31 16:34:35	ADXL345	mae	0.032288797
192	2022-03-31 16:34:36	ADXL345	mae	0.03035133
193	2022-03-31 16:34:37	ADXL345	mae	0.03259571
194	2022-03-31 16:34:38	ADXL345	mae	0.03350485
195	2022-03-31 16:33:56	ADXL345	avail	UNAVAILABLE
196	2022-03-31 16:34:25	ADXL345	avail	AVAILABLE
197	2022-03-31 16:34:25	ADXL345	condition	UNAVAILABLE
198	2022-03-31 16:34:26	ADXL345	condition	UNAVAILABLE
199	2022-03-31 16:34:25	ADXL345	execution	UNAVAILABLE
200	2022-03-31 16:34:26	ADXL345	execution	STOPPED
201	2022-03-31 16:34:41	ADXL345	Xacc	0.7741467
202	2022-03-31 16:34:42	ADXL345	Xacc	0.779404
203	2022-03-31 16:34:44	ADXL345	Xacc	0.8125154
204	2022-03-31 16:34:45	ADXL345	Xacc	0.8176314
205	2022-03-31 16:34:46	ADXL345	Xacc	0.8246659
206	2022-03-31 16:34:47	ADXL345	Xacc	0.8289092
207	2022-03-31 16:34:49	ADXL345	Xacc	0.6955928
208	2022-03-31 16:34:50	ADXL345	Xacc	1.177904

Result Grid
Form Editor
Field Types
Query Stats
Execution Plan

Figure 12 Result grid of stored data in MySQL workbench

Task 4.1

Capture the result grid as Figure 12 and then attach it to the report.

MySQL Workbench interface showing a query result for the 'robertclaud_lab7' table. The query is:

```
SELECT * FROM ME597Spring260.robertclaud_lab7
where timestamp > "2026-04-05 00:00:00.000000";
```

The result grid displays the following data:

id	timestamp	sensor	measurement	value
39061	2026-04-05 17:40:05.504294	ADXL345	Zacc	9.62100319385604
39062	2026-04-05 17:40:06.474235	ADXL345	Zacc	9.92799750707415
39063	2026-04-05 17:40:07.448554	ADXL345	Zacc	9.62633295969933
39064	2026-04-05 17:40:08.420123	ADXL345	Zacc	9.70027452414884
39065	2026-04-05 17:40:09.486946	ADXL345	Zacc	10.0251826133424
39066	2026-04-05 17:40:10.459740	ADXL345	Zacc	9.78097065503582
39067	2026-04-05 17:40:11.429263	ADXL345	Zacc	9.62511933604821
39068	2026-04-05 17:40:12.407242	ADXL345	Zacc	9.62930370973184
39069	2026-04-05 17:40:13.381405	ADXL345	Zacc	9.79494375810659
39070	2026-04-05 17:10:48.121906	ADXL345	mae	UNAVAILABLE
39071	2026-04-05 17:39:52.706913	ADXL345	mae	0.07652509
39072	2026-04-05 17:39:53.687131	ADXL345	mae	0.077070616
39073	2026-04-05 17:39:54.659134	ADXL345	mae	0.07508593
39074	2026-04-05 17:39:55.631030	ADXL345	mae	0.07569711
39075	2026-04-05 17:39:56.605051	ADXL345	mae	0.07686206
39076	2026-04-05 17:39:57.616150	ADXL345	mae	0.075619385
39077	2026-04-05 17:39:58.589764	ADXL345	mae	0.076317854
39078	2026-04-05 17:39:59.561311	ADXL345	mae	0.076394066
39079	2026-04-05 17:40:00.533475	ADXL345	mae	0.076223254
39080	2026-04-05 17:40:01.504509	ADXL345	mae	0.075497
39081	2026-04-05 17:40:02.481473	ADXL345	mae	0.07935726
39082	2026-04-05 17:40:03.561939	ADXL345	mae	0.08135727
39083	2026-04-05 17:40:04.534310	ADXL345	mae	0.08259369
39084	2026-04-05 17:40:05.504294	ADXL345	mae	0.08605615
39085	2026-04-05 17:40:06.474235	ADXL345	mae	0.1037093
39086	2026-04-05 17:40:07.448554	ADXL345	mae	0.1380787
39087	2026-04-05 17:40:08.420123	ADXL345	mae	0.2314805
39088	2026-04-05 17:40:09.486946	ADXL345	mae	0.26840332
39089	2026-04-05 17:40:10.459740	ADXL345	mae	0.27169818
39090	2026-04-05 17:40:11.429263	ADXL345	mae	0.19431774
39091	2026-04-05 17:40:12.407242	ADXL345	mae	0.18181121
39092	2026-04-05 17:10:48.122372	ADXL345	NULL	UNAVAILABLE
39093	2026-04-05 17:10:48.121106	ADXL345	NULL	UNAVAILABLE
39094	2026-04-05 17:10:48.122150	ADXL345	NULL	UNAVAILABLE
39095	2026-04-05 17:10:48.122070	ADXL345	condition	UNAVAILABLE
39096	2026-04-05 17:40:02.481473	ADXL345	condition	NORMAL
39097	2026-04-05 17:40:07.448554	ADXL345	condition	ABNORMAL
39098	2026-04-05 17:10:48.122296	ADXL345	execution	UNAVAILABLE
39099	2026-04-05 17:39:52.706913	ADXL345	execution	STOPPED
39100	2026-04-05 17:40:02.481473	ADXL345	execution	ACTIVE
39101	2026-04-05 17:40:14.352792	ADXL345	Xacc	1.04197250683718
39102	2026-04-05 17:40:15.420719	ADXL345	Xacc	0.879026007726011
39103	2026-04-05 17:40:16.390368	ADXL345	Xacc	0.901655634159201
39104	2026-04-05 17:40:17.368670	ADXL345	Xacc	0.883652630189487
39105	2026-04-05 17:40:18.346060	ADXL345	Xacc	0.873684790057017
39106	2026-04-05 17:40:19.325269	ADXL345	Xacc	0.892289186911775
39107	2026-04-05 17:40:20.297490	ADXL345	Xacc	0.865481394102033
39108	2026-04-05 17:40:21.362195	ADXL345	Xacc	0.843406564655041
39109	2026-04-05 17:40:22.341946	ADXL345	Xacc	1.07481170325166
39110	2026-04-05 17:40:23.314780	ADXL345	Xacc	0.85513988
39111	2026-04-05 17:40:24.302240	ADXL345	Xacc	0.932408938329095
39112	2026-04-05 17:40:25.277029	ADXL345	Xacc	1.06664619884283
39113	2026-04-05 17:40:26.246280	ADXL345	Xacc	0.558490913014341
39114	2026-04-05 17:40:27.321003	ADXL345	Xacc	0.389415638225191
39115	2026-04-05 17:40:28.292799	ADXL345	Xacc	0.385212402144708
39116	2026-04-05 17:40:29.262639	ADXL345	Xacc	0.366880918669748
39117	2026-04-05 17:40:30.244843	ADXL345	Xacc	0.400710807129258

4.2 Grafana dashboard

Finally, let's create a web-based dashboard using Grafana. Recall Lab8 activities. The information of the Grafana server domain and the port number is below. Log in the Grafana server in a web-browser of laptop.

- DNS: mepotrb16.ecn.purdue.edu
- Port: 3000
- Grafana web page URL: <http://mepotrb16.ecn.purdue.edu:3000/>
- Account (username): yourname (e.g., johndoe)

Go to 'Lab9 sample' dashboard in '0_SAMPLE' dashboard folder. This dashboard (Figure 13) is what you should finally create.

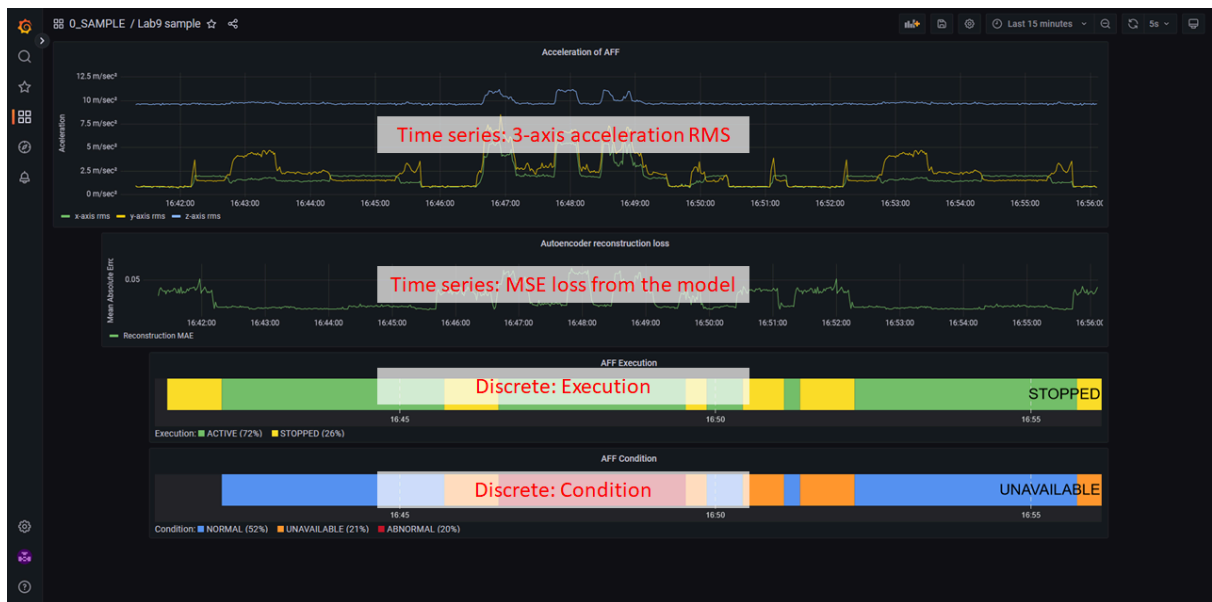


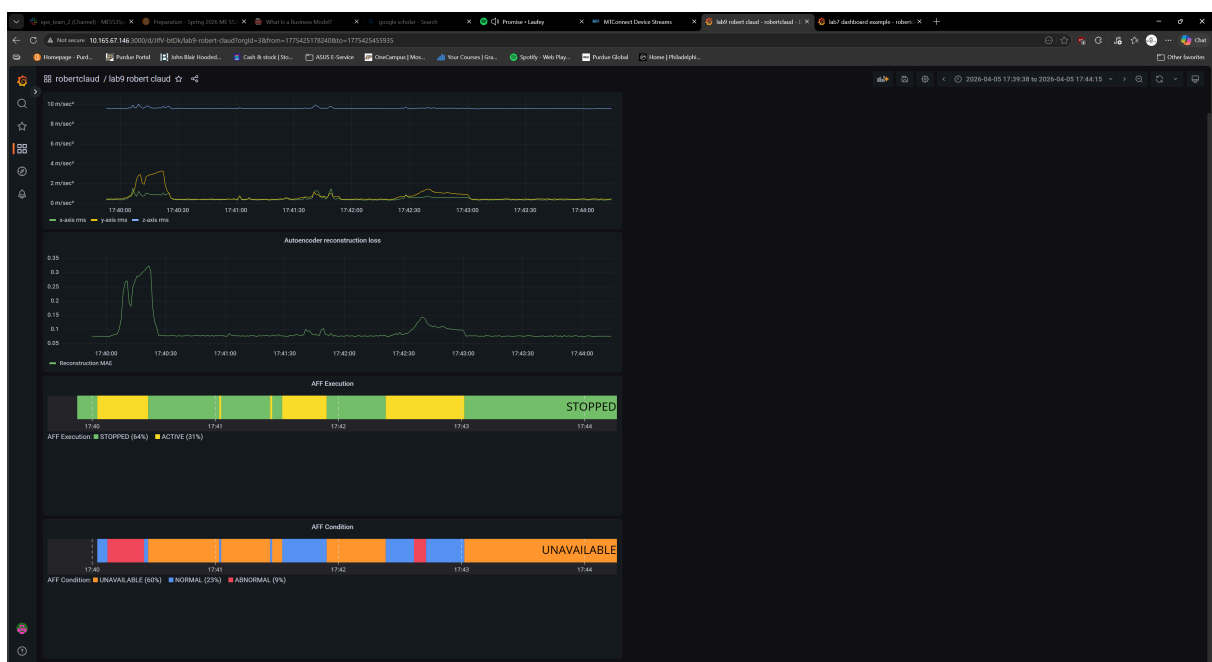
Figure 13 Sample Grafana dashboard (0_SAMPLE/Lab9 sample)

Task 4.2

Create a dashboard as Figure 13 in your folder created in Lab 7 and save it as the dashboard name of 'lab9 your name', e.g., 'lab9 john doe'.

Capture the dashboard and attach it to the report.

After creating the dashboard, confirm if the entire monitoring system works in real-time by refreshing the dashboard.



Lab9 Summary and Deliverables

Answer the following questions for your achievements

Q1. Please summarize Lab9.

In Lab 9 we deployed a previously trained autoencoder model to an edge device. In this case, the edge device was the Raspberry Pi with a connected ADXL345 accelerometer. From there, we developed an algorithm that could detect an rms threshold to determine if an axial flow fan was turned on or off. Using this threshold, once the AFF was determined to be turned on the autoencoder machine learning model was used to determine if the fan had a normal or abnormal operating condition. Using these conditions as well as the result of the ML model's reconstruction we dumped these values into an SQL database using the MTConnect Agent and Adapter and built a Grafana dashboard to monitor the results.

Q2. What skills did you have to develop to accomplish this project?

To accomplish this project, I needed to develop more of my ML skills and data processing.

Q3. What aspects of this project were the most beneficial for your learning?

The most beneficial aspects of this project were the fact that it was a real-world example that I could touch and interact with. It felt more real than a practice dataset to use machine learning on.

Q4. What challenges did you encounter in completing the project?

The challenge that I encountered was that the max, min and mae threshold values were wrong since the overall operating conditions from lab 8 was different from this lab.

Q5. How did you overcome the challenges or remedy the problems encountered?

To overcome this challenge, I quickly retrained a model under my current operating conditions and used new data to get the correct max, min and mae threshold values.

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